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LUNDS UNIVERSITET

FMN050 VT05: Numerical analysis for E/I

Numerisk Analys, Matematikcentrum

Project 1, selected answers

Last hand-in date: **2 May 2005 at 13:15**

Aim: To study interpolation in an applied problem; the role the basis functions play for the accuracy (conditioning); why it is important for modelling purposes to keep the degree of the interpolation polynomial low.

Grading: Accounts for 10p of the course's total 50p (5p required for passing grade). **NOTE:** Make sure that your report has your name, personnummer, section and your class year, e.g. E03.

Reports: **The project must be solved and reported individually.** You are encouraged to discuss the problems with your fellow students, but you must construct your own MATLAB programs yourself, run them, interpret the results, and report on your findings in your own report to be graded.

The projects are part of the examination, and your approved and graded project reports remain valid for one year.

Consider the population data presented in the Table below (from Heath, Scientific computing, 1997, p. 238).

Year	Population
1900	76,212,168
1910	92,228,496
1920	106,021,537
1930	123,202,624
1940	132,164,569
1950	151,325,798
1960	179,323,175
1970	203,302,031
1980	226,542,199

The eighth degree polynomial $p_8(t)$ that interpolates the nine data points can be represented by a basis $\{\phi_0(t), \phi_1(t), \ldots, \phi_8(t)\}$. Consider the following basis functions:

$$
\begin{array}{ll}
a) & \phi_j(t) = t^j \\
b) & \phi_j(t) = (t-1900)^j \\
c) & \phi_j(t) = (t-1940)^j \\
d) & \phi_j(t) = ((t-1940)/40)^j
\end{array}
$$

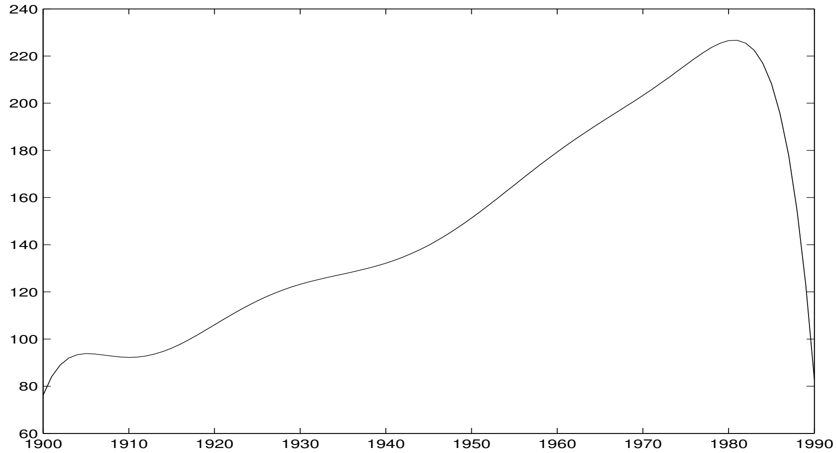

Figure 1: *Interpolation polynomial*

Problem 1. The 2–norm condition numbers are a) $7.9 \cdot 10^{36}$ (hopeless for numerical computations, this basis must be discarded); b) $6.1 \cdot 10^{15}$ (much better, but still hopeless); c) $9.3 \cdot 10^{12}$ (fair enough, one might get a few digits' accuracy, but one can do far better, as d) shows, where $\kappa \approx 1.6 \cdot 10^3$; here it is finally possible to get some reasonable results. It doesn't matter very much if you choose another norm than the 2–norm.

Problem 2. Represent the polynomial $p_8(t)$ in the best-conditioned basis found in the previous part. Evaluate $p_8(t)$ at one-year intervals and plot $p_8(t)$ together with the interpolation points.

Problem 3. You get a poor prediction for 1990, 82.75 million, which is nowhere near the correct value of 248.71 million. The high degree polynomial "runs away" outside the data interval.

Problem 4. You get the same plot if you use Newton's interpolation basis. This is because the interpolating polynomial is unique; the Newton basis is only a different representation of the same polynomial (although with a different condition number).

Problem 5. Here one needs to try out different degrees of the polynomial for the least squares fit. Below are the plots for degree 4 and 5. As degree 4 has a smoother behaviour near 1990, we prefer using that degree. The residual is small, and the predicted value 243.1 is in reasonably good agreement with exact data.

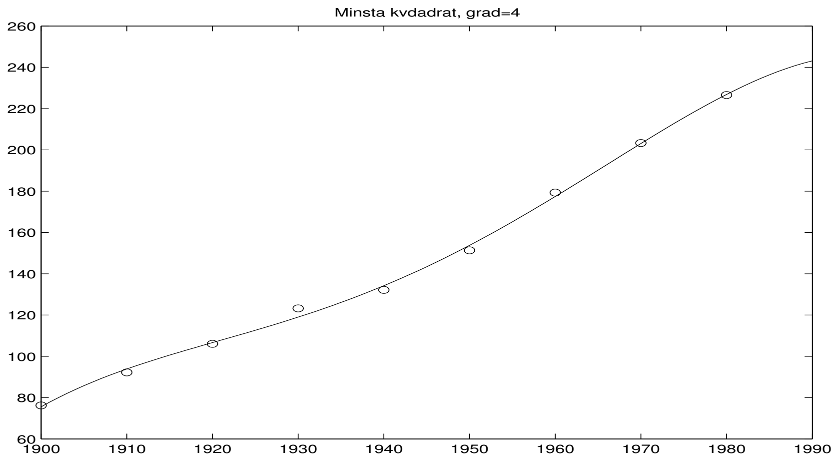

Figure 2: *Least squares solution, degree 4*

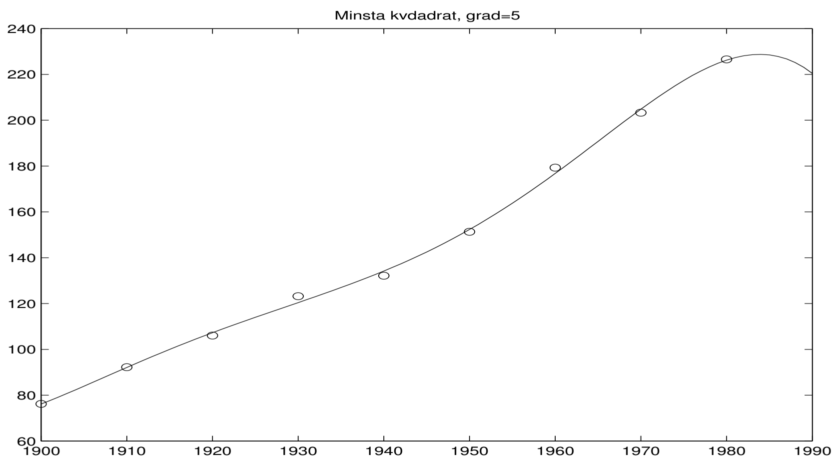

Figure 3: *Least squares solution, degree 5*